(RT2-1) Investigating a Plasmid and Recombination-Based System for Genome Editing in *Enterobacter ludwigii*

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Background

- In hospital plumbing systems, P-traps serve as breeding grounds for microbes that spread upwards to reach drain openings in sinks and are subsequently dispersed throughout the surrounding environment[1]
- Many plumbing-associated microbiomes harbor pathogens, including those that are multidrug-resistant[2]
- Transmission of such dangerous microbes within the hospital setting poses a clear public health concern, especially considering hospitals' relatively high density of patients who are immunocompromised, elderly, etc.
- A common strategy to remove unwanted sink colonization is the application of disinfectants and detergents
- This approach is rarely effective beyond the short run and can contribute to the emergence of microbial resistance[3]
- One promising alternative is the development and use of probiotic-based systems[4,5]
- *Enterobacter ludwigii* is a Gram-negative, opportunistic bacteria capable of colonizing a variety of environments, with high abundance in the sink built environment
- Minimal research has been conducted with sink-adapted *E. ludwigii*

Objectives

General Objectives

1) Improve basic, biological understanding of *E. ludwigii*
2) Develop the tools and protocols necessary to genetically engineer *E. ludwigii* to possess more probiotic and/or antimicrobial characteristics

Summer Project Objectives

1) Computational: develop programs to identify possible locations for insertion within a genome and provide preliminary information regarding potential downstream effects of a given insertion
2) Wet lab: conduct foundational genome editing experiments – focused on utilizing a designed plasmid and recombination system

Literature Cited

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3. Parkes, L. O., & Hota, S. S. (2018). Sink-Related Outbreaks and Mitigation Strategies in Healthcare Facilities. *Current Infectious Disease Reports*, *20*(10). https://doi.org/10.1007/s11908-018-0648-3
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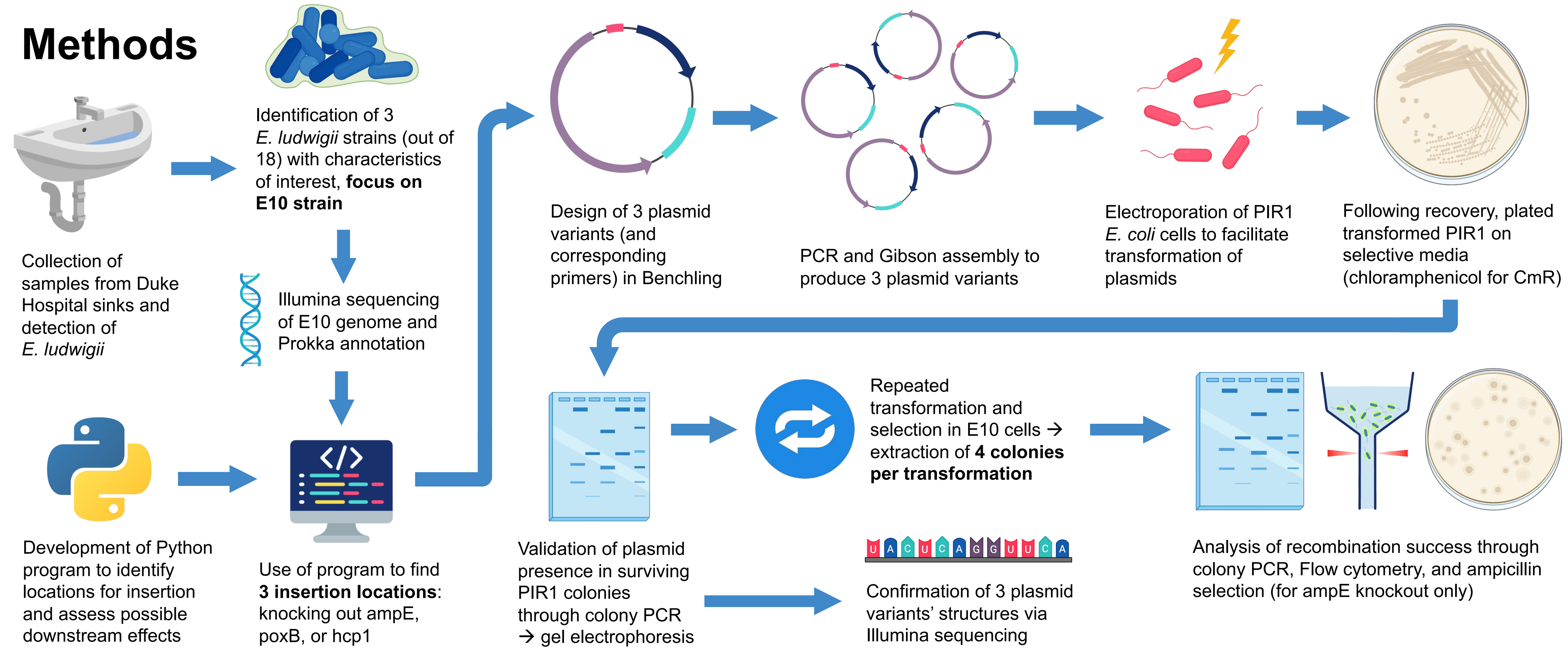
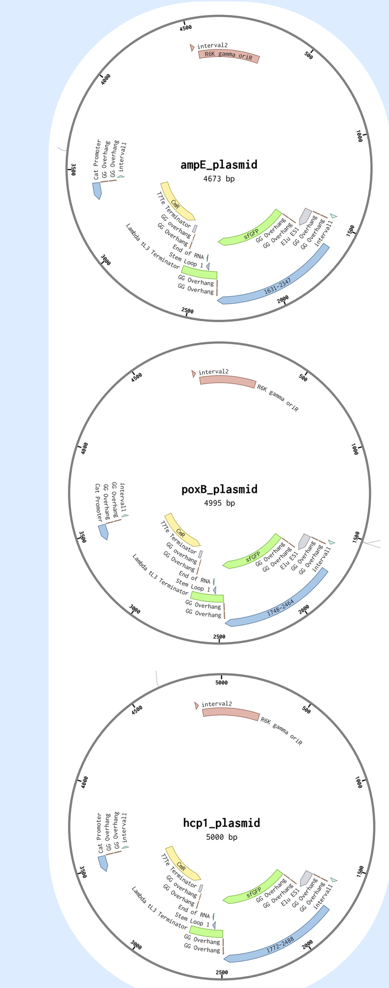
Methods



Plasmid Design

- 3 plasmid variants, one for each desired insertion location (ampE, poxB, hcp1)
- All 3 contain: insert (GFP and CmR genes), origin that is functional in *E. coli* but **not** in E10
- Each plasmid variant also contains appropriate homologous arms
- Following successful recombination, E10 cells would fluoresce (due to GFP) and be chloramphenicol resistant (due to CmR)



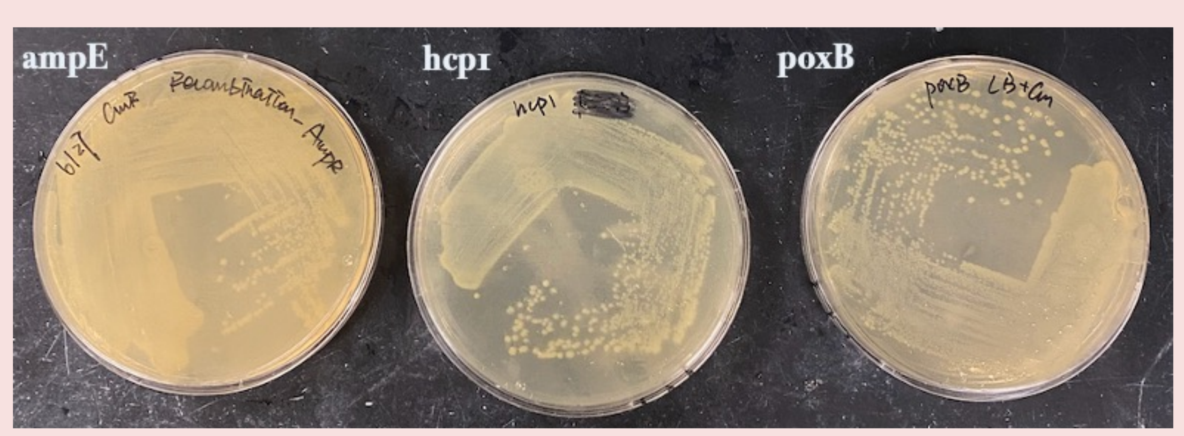
Results

PART 1: Development of Computational Tools

- Successfully created a program with two useful functions:

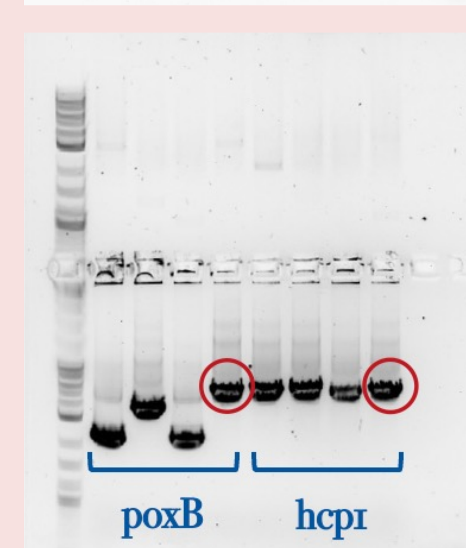
1) Identification of potential locations for insertion within an (annotated) genome
   a) Known gene(s) that user wishes to knock out
   b) Intervals that are *not* associated with any coding sequences
2) Compares two annotated genomes (pre- and post-insertion) and returns information regarding genes that may be lost or gained due to a given insertion taking place
   a) Creates summarizes of changes in both known and hypothesized genes

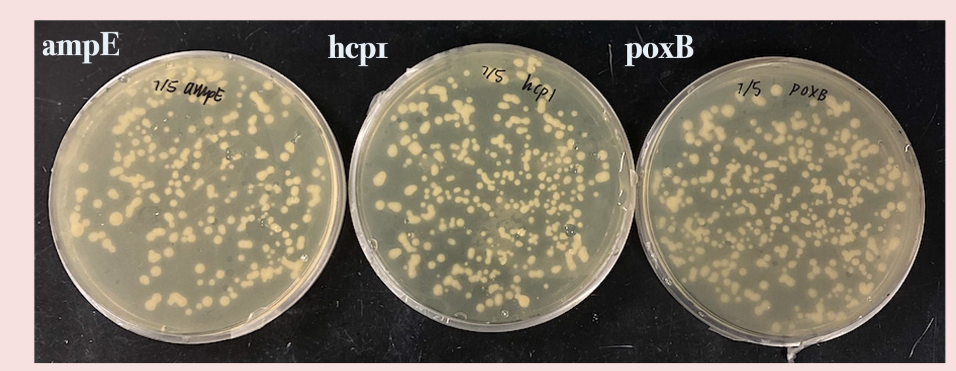
- Used program to locate and retrieve sequence information about the three desired insertion locations
- Relevant Python files and appropriate documentation published on GitHub for future access/use



PART 2: Plasmid Design, Production, and Transformation

- Successfully designed and produced three plasmid variants, verified plasmids' structures through Illumina sequencing
- Successful transformation of each plasmid type into PIR1, as evidenced by survival on selective media (chloramphenicol for CmR) and bands of appropriate length following colony PCR
- Likely achieved proper transformation of plasmids into E10, based on seeing appropriate voltage and time values during electroporation (5.6 kV, 5.8 ms)

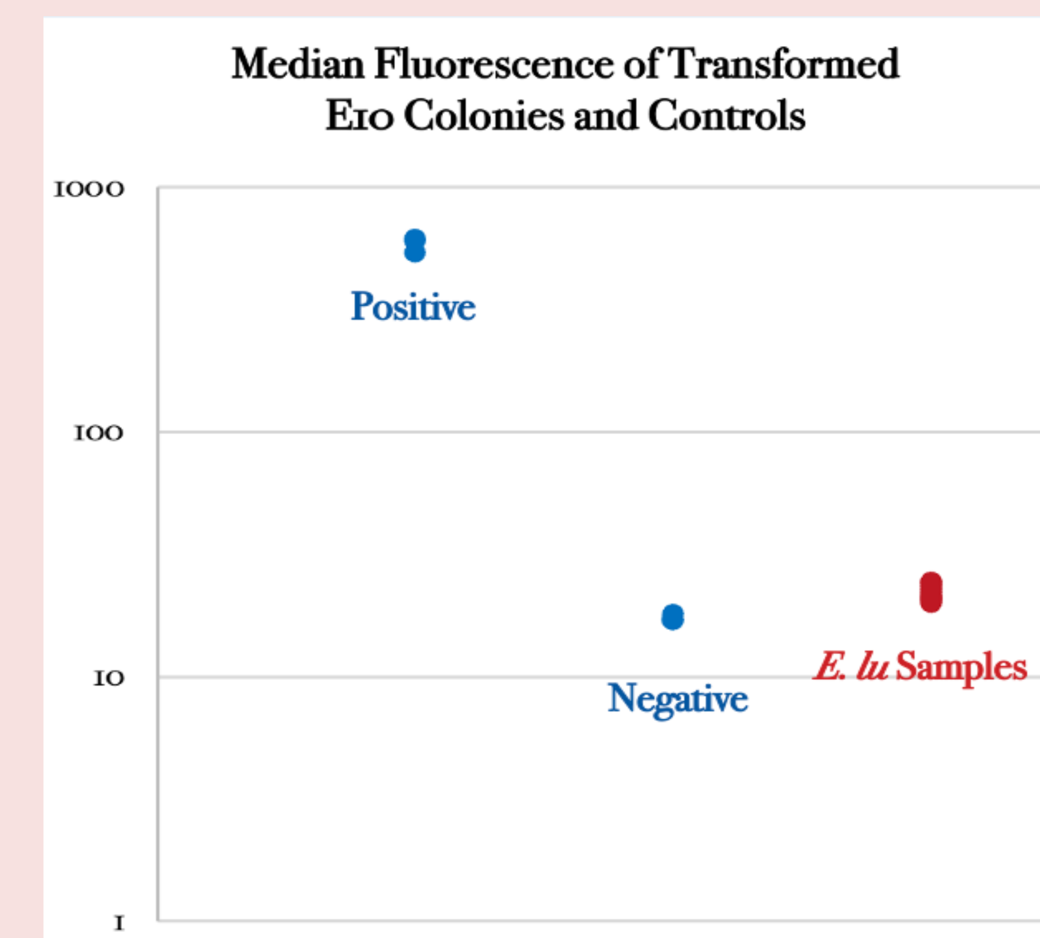


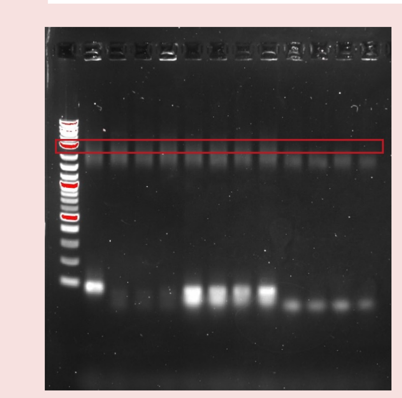


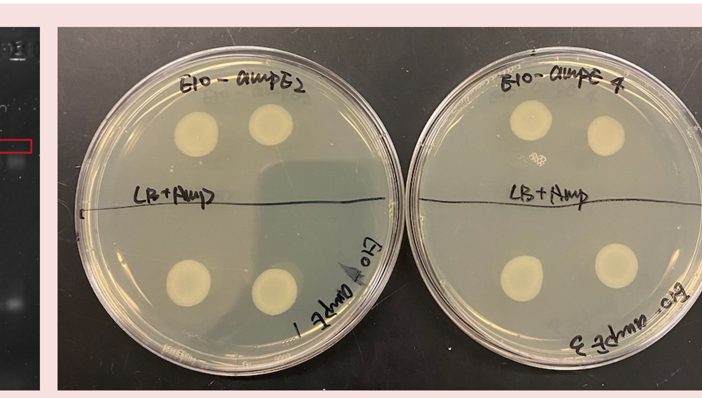
PART 3: Recombination in E10

- All 12 E10 colonies sampled (4 per transformation) were false positives
- Despite passing selective marker test for CmR, none had integrated the insert into the genome

- Median fluorescence levels of samples (obtained through Flow cytometry) closely matched those of negative controls → indicates lack of GFP gene in genome
- Gel electrophoresis run with colony PCR products yielded no bands of appropriate length (desired length of ~3 kb indicated in red) → indicates lack of insert in genome
- Intended ampE knockout survived on ampicillin-containing media → indicates ampE was *not* removed from genome and replaced with insert







Future Directions

Immediate

- Continue using program to locate locations for insertion and analyze different locations' relative desirability
- Repeat experiment, tweaking certain aspects of protocol (ex: increase length of homologous arms in plasmids, sample more transformed *E. ludwigii* colonies)
- Explore CRISPR technology as another method of genetic engineering of *E. ludwigii*

Long Term

- Determine useful mechanisms for *E. ludwigii* to employ within sink microbiome
- Identify and harness specific functional genes to increase *E. ludwigii* robustness and probiotic/antimicrobial characteristics

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